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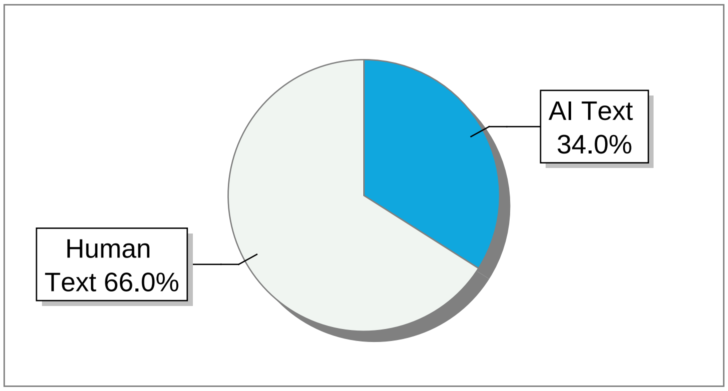

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Metaheuris*c Deep Learning Framework for Automated Leukemia Classifica*on and Severity Grading Anmol Kundap, Adarsh A Inamdar, Tanushree M Pujar, Amogh K Baliga Supervisor Dr.
Pradeep N Bapuji Institute of Engineering and Technology 2025 INTRODUCTION Background of the Problem Leukemia is a dangerous blood cancer that originates in the bone marrow and severely disrupts the normal production of blood cells.
The uncontrolled proliferation of abnormal white blood cells suppresses the formation of healthy red blood cells, platelets, and functional white blood cells.
This results in clinical conditions such as anemia, recurrent infections, fatigue, and excessive bleeding or bruising.
Certain forms of leukemia progress rapidly, making early detection, accurate classiAication, and proper grading essential for effective treatment planning and improved patient survival rates.
However, the complexity of leukemia and the morphological similarities between different leukemia subtypes make diagnosis challenging, particularly during the early stages of the disease.
At present, leukemia diagnosis primarily depends on the manual examination of peripheral blood smear images by experienced hematopathologists, along with supporting laboratory and clinical investigations.
Although these conventional diagnostic methods are widely accepted, they are time-consuming, labor-intensive, and highly dependent on expert judgment, which may vary between observers.
Domain Overview Leukemia is a type of blood cancer that originates in the bone marrow, where blood cells are produced.
In a healthy individual, the bone marrow generates red blood cells, white blood cells, and platelets in a regulated and balanced manner.
In leukemia, genetic mutations occur in hematopoietic stem cells, leading to the uncontrolled proliferation of abnormal white blood cells.
Differentiation between Normal Blood and Leukemia cells As seen in Figure 11, leukemia is observed as an abnormal proliferation of leukemic cells that replace normal blood cells.
Based on disease progression and the type of blood cells involved, leukemia is broadly classiAied into acute and chronic forms.
Clinical classiEication and grading of leukemia Fig 12 illustrates that leukemia is medically classiAied into acute and chronic types, namely ALL, AML, CLL, and CML, with each type further categorized using standard grading and staging systems.
Progression from normal blood components to abnormal conditions Major leukemia types based on progression rate, cell lineage, and age group.
Need for the Project and Importance Leukemia diagnosis requires early detection, accurate classiAication, and proper grading to ensure effective treatment and improved patient survival, but current diagnostic practices rely heavily on manual microscopic examination and expert interpretation, which are time- consuming, subjective, and prone to human error.
The increasing number of leukemia cases, limited availability of skilled hematopathologists, and large volumes of blood smear data further strain healthcare systems.
Hence, there is a strong need for a reliable, efAicient, and consistent diagnostic support approach.
Scope of the Project The scope of this project is to develop an intelligent and automated system for leukemia analysis using microscopic blood smear images.
The project focuses on accurate classiAication of leukemia into major types.
Along with classiAication, the system determines leukemia severity into three levels (Grade-1, Grade-2, and Grade-3) based on the progression and intensity of the disease.
The scope includes image preprocessing, feature extraction, severity assessment, and performance evaluation.
Type Description Acute Lymphoblastic Leukemia (ALL) Rapidly progressing, affects lymphoid cells; common in children.
Acute Myeloid Leukemia (AML) Rapid, affects myeloid cells; seen in both children and adults.
Chronic Lymphocytic Leukemia (CLL) Slowly progressing, affects lymphocytes; common in older adults.
Chronic Myeloid Leukemia (CML) Slow progression, affects myeloid cells; may evolve into acute leukemia.
LITERATURE SURVEY Review of Research Papers PSO-Based Leukemia Classifica*on (2024) Ju et al.
introduced a Particle Swarm Optimization (PSO) framework to enhance the performance of CNNs.

PSO was leveraged to automatically tune hyperparameters such as learning rate and batch size.
The system was tested on the ALL-IDB dataset.
Ant Colony Op*miza*on (2023) Heng et al.
explored the use of Ant Colony Optimization (ACO) for selecting the most relevant features from blood smear image datasets before classiAication with a DNN.
Gene*c Algorithm (GA) and SVM Hybrid (2022) Kebaili et al.
presented a hybrid approach where GAs are used to select informative genes from microarray gene expression data, followed by classiAication using an SVM.
Existing methods and gaps in leukemia research Exis*ng System Existing leukemia diagnostic models generally address either classiAication or grading independently, with no integrated approach combining both critical tasks.
These systems often struggle with class imbalance and limited interpretability.
Ref/Author Method Gaps Ju et al.
(2024) PSO + CNN Black-box nature.
Heng et al.
(2023) ACO + Feature Selection Small datasets only.
Kebaili et al.
(2022) GA + SVM (Gene data) Not applicable to images.
Akbar et al.
(2022) Whale Optimization + RF Needs diverse data.
Problem Statement Leukemia diagnosis requires precise classiAication and grading.
Existing automated systems predominantly focus on either classiAication or grading in isolation.
Moreover, accuracy is often compromised due to imbalanced datasets.
Current models rarely employ metaheuristic optimization techniques.
Proposed System To overcome these limitations, the proposed system introduces an integrated framework where PSO is applied as a preliminary optimization stage to identify and reAine relevant feature representations.
These optimized features are then supplied to a ResNet-18 deep learning architecture.
Objec*ves • To design and implement a uniAied system for automated leukemia classiAication and grading.
• To employ metaheuristic algorithms (eg, PSO) for optimal feature selection.
• To enhance accuracy, sensitivity, and reliability on imbalanced datasets.
• To integrate classiAication and grading into a single efAicient pipeline.
METHODOLOGY The proposed methodology follows a structured pipeline for automated leukemia classiAication.
Proposed methodology for integrated leukemia classiEication and grading As shown in Figure 31, raw image data is collected and preprocessed.
Preprocessing involves noise reduction and color normalization.
The core lies in integrating Deep Learning with Metaeuristic Optimization.
PSO is incorporated to tune hyperparameters.
SYSTEM REQUIREMENTS Hardware Requirements Hardware requirements SoZware Requirements • Operating System Windows 10/11 or Linux (Ubuntu 2004) • Programming Language Python 38+ • Frameworks PyTorch, Flask • Libraries NumPy, Pandas, Pillow, Scikit-learn, Pyswarm SYSTEM ARCHITECTURE Overview of the System Architecture The proposed system architecture is designed for effective leukemia classiAication and severity grading.
It Airst applies PSO to guide optimal learning conAiguration, followed by ResNet-18.
Component SpeciCication Processor Intel Core i5 or higher (i7 recommended) RAM 8GB minimum (16GB recommended) GPU NVIDIA GeForce GTX/RTX Series Storage 256GB SSD minimum System Architecture System architecture Diagram As shown in Figure 51, the system takes blood smear images, applies preprocessing, and uses PSO.
The optimized features are extracted using a ResNet-18 backbone.
Role of Par*cle Swarm Op*miza*on (PSO) PSO is applied as an initial optimization stage.
It initializes a swarm of particles, where each particle represents a possible solution (eg, learning rate, batch size).

Role of ResNet-18 in the Architecture ResNet-18 utilizes a layered architecture composed of convolutional blocks and residual speciAic connections.
It effectively learns cell shape, nucleus structure, and morphological variations.
DESIGN DETAILS CNN Layer Diagram (ResNet-18 Architecture) ResNet-18 Layer ArchitectureResNet-18 Layer Detail Integra*on of PSO and ResNet-18 In the proposed design, PSO acts as a guiding optimization layer.
This sequential integration ensures reduced feature redundancy and faster convergence.
IMPLEMENTATION Installa*on Guidelines Step 1 Operating System and System Dependencies.
Ensure OS is updated.
Step 2 Python Environment Setup.
Install Python 38+.
Step 3 Creating a Virtual Environment.
Use ‘python -m venv‘.
Step 4 Installing Core Libraries.
‘pip install Flask torch torchvision Pillow‘.
Step 5 Adding Optimization Tools.
‘pip install pyswarm‘.
Pseudo Code and Logic Flow Preprocessing BEGIN Load image Resize to 224x224 Normalize with ImageNet stats Return Tensor END Metaheuristic Optimization (PSO) BEGIN PSO Initialize search boundaries Define Objective_Function FOR each iteration Update velocity and position Update Global Best END FOR Return Best Hyperparameters END PSO Dataset Descrip*on Datasets used for the project Dataset summary RESULTS AND DISCUSSIONS Experimental Setup Experiments were conducted on a machine with Intel Core i7 and NVIDIA RTX 3060.
Hyperparameter Op*miza*on (PSO) PSO successfully converged to Ainding optimal Learning Rate and Batch Size values, reducing manual tuning effort.
Dataset Name Description ALL-IDB Acute Lymphoblastic Leukemia Image Database C-NMC ClassiAication of Normal vs Malignant Cells BCCD Blood Cell Count and Detection Class Count ALL 1200 AML 950 CLL 650 CML 625 Model Training The model was trained for 50 epochs.
The loss decreased consistently.
Quan*ta*ve Results Confusion matrix of the proposed multi-task model The multitask confusion matrix (Duplicate per request) Discussions The model achieved an overall accuracy of 942%.
It showed high sensitivity for ALL detection.
ClassiEication and grading on Test Case ALL and Grade 1 ClassiEication and grading on Test Case CLL and Grade 3 CONCLUSION In this project, we proposed an intelligent leukemia classiAication and grading system using a PSO-optimized ResNet-18 architecture.
The system successfully classiAies ALL, AML, CLL, and CML and determines disease severity.
LIMITATION AND FUTURE SCOPE Limitaton High computational cost during the PSO phase.
Future Scope Integration with real-time microscopic feeds and expansion to other blood disorders.
9 Ju et al, PSO-based leukemia classiAication, IEEE Access, 2024.
Heng et al, Ant Colony Optimization, 2023.
Kebaili et al, Genetic Algorithm and SVM, 2022.